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Weight Loss and Food Spending

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ABSTRACT

US consumers are increasingly focused on weight management. However, the economic impacts of this on the US food industry are not well understood. Thus, we estimate the effects of weight loss attempts on the consumption of and expenditures on a collection of 35 food groups. We obtain weight loss history and food consumption data from the National Health and Nutrition Examination Survey and Purchase to Plate Suite. Attempting weight loss is an endogenous choice and, as such, we use matching methods to reduce confounding. Matched samples and weights are used in lower-limit censored outcome models of food consumption and expenditures. Nationwide weight loss attempts yield increases in daily expenditures on non-citrus fruits, non-starchy vegetables, and tomatoes of \$17.8 million. Daily, national expenditures on processed grain products and soft drinks decrease by \$18.8 million and \$7.3 million, respectively. Food manufacturers and retailers can use these results to develop business strategy and mitigate risks associated with changing consumer behavior related to weight management. These findings can also guide public health and nutrition policies, directly speaking to how the composition of diets changes as consumers attempt weight loss.

JEL Classification: D12, Q10

1 | Introduction

US citizens are increasingly concerned with weight management. However, little is known about how this impacts the US food industry. Fundamental concepts in behavioral economics suggest contrasting impacts of weight loss on food consumption and expenditures. Various studies find that food deprivation increases the absolute or relative reinforcing value of food (Epstein et al. 2003; Raynor and Epstein 2003), or how hard an individual is willing to work to gain access to food [placed in relative terms if measured when different types of food or nonfood items are available] (Epstein and Leddy 2006; Hill et al. 2009). Thus, individuals who are attempting weight loss may not adhere to their diet and consume more, potentially unhealthy foods. Conversely, present bias (i.e., the simultaneity effect) reflects an inclination to choose a smaller, present reward to a larger, later reward but to reverse the preference when both rewards are equally delayed (Chakraborty 2021). Weight loss attempts may serve to reduce present bias as they represent a conscious and effortful

commitment to longer-term health benefits. That is, relative to other individuals, those attempting to lose weight may be less inclined to consume food in the short term to ensure reduced weight and improved health outcomes in the future. Further, food products exhibit notable heterogeneity in their caloric densities and nutrient compositions, which likely influences which foods are preferred when weight loss is attempted. Collectively, these concepts motivate the objective of this study, which is to quantify the impacts of intentional weight loss attempts on both the consumption of and expenditures on various food groups.

A host of government, industry, and academic sources suggest that weight management is increasingly important to US citizens. In 2024, 27 percent of US adults reported that they were seriously attempting to lose weight, reflecting a seven-percentage point increase from the prior year (Brenan 2024). Some portion of these individuals may be electing to lose weight through traditional strategies, like physical exercise, with the US Bureau of Labor Statistics (2024) estimating a 1.8 percentage point increase in

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the share of US citizens participating in sports, recreation, or other exercise from 2019 to 2023 and Planet Fitness [the largest US gym chain] (2024) reporting a 6.8 percent compound annual growth rate in memberships over the same period. The growing rate of US adults who are trying to lose weight may also reflect the recent adoption of glucagon-like peptide-1 (GLP-1) receptor agonists, which suppress users' appetites and result in lower total caloric intake (Latif et al. 2024; Mayo Clinic 2024). Early public polling efforts indicate that between three and six percent of US adults are using the medications (Montero et al. 2024; Witters and Maese 2024) and that most users have changed the types of foods that they consume (Roe 2025). Collectively, trends in US citizens' focus on weight management may have meaningful impacts on the effectiveness of public health campaigns and policies, and on economic outcomes experienced by the nation's food producers. These potential impacts are not well understood, motivating this work.

We fulfill our stated objective using observational data on US adults' weight loss history and food consumption from the National Health and Nutrition Examination Survey (NHANES). Attempting weight loss is an endogenous choice and, accordingly, we implement matching techniques to balance covariates between consumer groups that do and do not attempt to lose weight. We use the matched samples and weights to estimate a series of lower-limit censored food consumption and expenditure models. The effects of weight loss attempt on consumption and expenditures are then estimated using a doubly robust standardization procedure. We find that intentional weight loss yields an increase in daily consumption of, primarily, single-ingredient foods such as eggs, non-citrus fruits, non-starchy vegetables, nuts, poultry, and tomatoes. Conversely, the consumption of dairy, potatoes, processed grain products, and soft drinks is reduced. These effects reflect increases in daily, national expenditures on produce of \$17.8 million and reductions in daily, national expenditures on processed grain products and soft drinks of \$18.8 million and \$7.3 million, respectively.

Our research contributes to two separate literatures that are relevant to our understanding of the importance of weight management in the US food industry. First, we contribute to the food marketing and industrial organization literature by providing empirical estimates of the impacts of weight loss attempts on food purchasing behavior. Our findings are important in this regard because they improve firms' ability to develop marketing and pricing strategies to mitigate the risk of changing consumer behavior. Second, we expand on public policy literature on weight loss and food choice by evaluating the impacts of weight loss attempts across a broader collection of foods that better reflect the full choice set available to US consumers and by considering the economic impacts (i.e., expenditures) rather than just consumption. This contribution is important in that our results speak to how the composition of diets change as consumers attempt to lose weight. Further, these results provide improved understanding on which food groups "win" or "lose" (in terms of total expenditures) when consumers become more focused on weight management. That is, we provide insights on which sectors in the US food industry are likely to benefit from observed increases in the number of consumers attempting to lose weight (Brenan 2024) and which sectors are made worse off.

The remainder of this study is as follows. First, we provide a brief synthesis of prior research assessing the relationship between weight loss and food consumption. We then overview the survey data on weight loss history, food consumption, and food expenditures that informs our analysis. Following this, we discuss our empirical approach. Primary results, heterogeneity assessments, and important considerations are then presented. We conclude with a discussion on policy implications of this work and future areas of research related to the topic.

2 | Synthesis of Prior Work

A small body of literature examines the effects of weight concerns and weight loss intentions on food consumption. Glanz et al. (1998) assesses the importance of various factors, including weight control, on dietary choices in a national survey of US adults. They find that the importance placed on controlling weight is positively associated with consumption of fruits and vegetables and negatively associated with consumption of fast food and cheese. Fan and Jin (2014) additionally use survey-based data to estimate differences in eating behaviors between US citizens that vary in their obesity status and in the strength of their weight loss intentions. They find that overweight and obese individuals are less likely to meet federally-recommended fruit consumption levels relative to their normal-weight peers, and that overweight and Class I and II obese individuals are less likely to meet recommended vegetable consumption levels. However, the authors do not quantify the association of weight loss intention itself with food consumption.

Relative to those studies, our work is more adjacent to prior efforts estimating the association of actual weight loss attempts with food consumption. In a cross-sectional analysis of UK adults, Wardle et al. (2000) find that those who are trying to lose weight are more likely to report restricting their consumption of fats and sugars than those who are not, but that no difference exists in consumption of fruits or vegetables. Likewise using cross-sectional analysis of survey responses, Bleich and Wolfson (2014) quantify the association of attempting to lose weight with consumption of sugar-sweetened beverages and snacks among US adults. The authors find that adults trying to lose weight consume 50, 25, and 30 calories less per day from sugar-sweetened beverages, salty snacks, and sweet snacks, respectively, relative to adults not trying to lose weight. Importantly, these efforts focus on a limited group of food products that can generally be classified as "healthy" (i.e., fruits and vegetables) or "unhealthy" (e.g., sweet snacks). Further, this prior work does not consider the *economic* impacts of weight loss attempts, focusing solely on food consumption rather than expenditures, and does not consider that underlying, secondary characteristics of consumers (e.g., sex, age, income) may drive purchasing decisions rather than attempting weight loss.

3 | Materials and Methods

3.1 | Data

We use data from the 2011–2012 through 2017–2018 cycles of NHANES, which includes 70,190 participants in total. NHANES

is a nationally-representative survey that is designed to provide information on the health and nutritional status of the civilian, non-institutionalized US population (Chen et al. 2020) and is conducted by the National Center for Health Statistics (NCHS) branch of the Centers for Disease Control and Prevention. NHANES data takes a pooled cross-sectional structure, with a different subset of the US population surveyed in each 2-year cycle. As part of the NHANES project, a 24-h dietary recall is obtained during participants' visits to a mobile examination center (MEC) with a second 24-h dietary recall obtained for most participants via telephone within 3–10 days following the MEC examination (Zipf et al. 2013). Additionally, sociodemographic characteristics and participants' history of attempted weight loss are obtained during in-home interviews.

During the MEC examination, NHANES participants are prompted to recall, at the product level, (i) all foods and beverages they consumed the prior day (i.e., midnight to midnight), (ii) at what time of day the food or beverage was consumed, (iii) the name of the eating occasion (i.e., breakfast, lunch, dinner, etc.), and (iv) where the food or beverage was obtained and whether it was consumed at home or away from home (National Center for Health Statistics 2024a). Additionally, participants provide an estimate of the volume of consumption for each food or beverage using a standard set of measuring guides provided by survey administrators (Centers for Disease Control and Prevention 2024b). Since 2001, food and beverage consumption reported in the dietary component of the NHANES survey project has been categorized into distinct food product codes according to the USDA's Food and Nutrient Database for Dietary Studies [FNDDS] (Ahluwalia et al. 2016). Further details on data collection methods are readily available in prior reports (Ahluwalia et al. 2016; Dwyer et al. 2003; Food Surveys Research Group 2024; National Center for Health Statistics 2024a). Importantly, Ahluwalia et al. (2016) and Dwyer et al. (2003) provide overviews of the limitations of NHANES dietary data, along with other analytical considerations. These limitations include the potential underreporting of total caloric intake among certain participant groups and large intraindividual day-to-day variations in dietary intake that make individual-level analysis inappropriate. However, large sample sizes, high response rates, and the 24-h recall (requiring short-term memory, rather than long-term memory) are noted as strengths that make the NHANES data useful in capturing mean effects and informing policy at the population level.

For statistical analyses, we aggregate participants' reported first-day dietary intakes to the food subgroups defined in the FNDDS (Food Surveys Research Group 2024; Love et al. 2020; Mishra et al. 2017; Pikosky et al. 2022; Willits-Smith et al. 2023). We then present results for food subgroups that are reported sufficiently frequently to allow for estimable regression models and trustworthy parameter estimates. Importantly, the "Nonalcoholic beverages" subgroup consists of a host of products that vary widely in their characteristics (e.g., caffeination, carbonation, sugar content, etc.), which yields nonsensical statistical estimates in our weight loss-focused application. As such, we assess consumption and expenditures only for products in this subgroup that are defined as "soft drinks" (i.e., soda, pop) and are not the diet, or sugar-free versions. Appendix Table A1 provides an

overview of the 2017–2018 FNDDS food code, which includes nine major food groups and their subgroups.

Prices for NHANES-reported foods are obtained from the USDA Economic Research Service's (ERS) Purchase to Plate National Average Prices dataset (2024). Detailed documentation of the Purchase to Plate project is available in prior reports (Carlson et al. 2019, 2020, 2022). Briefly, universal product codes (UPCs) found in (now Circana) Information Resources, Inc.'s (IRI) retail grocery scanner data are mapped to FNDDS food codes found in the NHANES dietary data using a combination of semantic, probabilistic, and manual matching. The researchers' matching process requires that (i) 95 percent of IRI-recorded food sales must be matched to an FNDDS food code and (ii) the error rate must be less than five percent for any IRI product category (Carlson et al. 2022). After matching, various conversion factors (documented elsewhere) are applied to account for differences in IRI and FNDDS data (i.e., purchase volumes vs. consumption volumes, foods prepared with recipes and multiple ingredients). The total edible weight sold of an FNDDS food is then the summation of edible weights sold across all UPCs that are matched to the FNDDS food (Carlson et al. 2020). The price per edible kilogram of the FNDDS food is then the total sales of the same UPCs divided by the total edible weight sold. These prices reflect national averages observed during the specific NHANES cycle and are not adjusted for seasonality. We multiply these product prices by the reported intake of each food item to determine NHANES participants' expenditures. Expenditures are then aggregated to the FNDDS food subgroups (Appendix Table A1).

Beyond the dietary recall, NHANES participants complete an in-home questionnaire, which asks "*During the past 12 months, have you tried to lose weight?*" (National Center for Health Statistics 2024a). Pursuant to our stated objective, we leverage this question to quantify the impacts of intentional weight loss attempts on food consumption and expenditures. Further, NHANES participants are prompted to indicate which weight loss strategies were implemented [e.g., switched to lower-calorie foods, exercised, skipped meals, etc.] and are able to report multiple strategies. Many of these strategies are reported infrequently (e.g., taking weight loss medications) or are otherwise reported simultaneously with other strategies, precluding estimation of their impact on food consumption and spending behavior. Thus, this study focuses on weight loss attempts in general.

The analytical sample is restricted to participants aged 18 years and older with complete information for key sociodemographic characteristics and weight loss history, and who provide a reliable dietary recall (i.e., all prior-day foods are listed [midnight to midnight], the time and eating occasion are provided, and a detailed description of the food and amount eaten is provided). This leaves 16,665 participants for further analysis. The food consumption and expenditures of the usable sample are described in Table 1, distinguishing between participants who attempted to lose weight in the prior year and those who did not. Appendix Table A2 provides further reporting of the sample's sociodemographic characteristics.

Briefly, 38 percent of our sample indicated that they attempted weight loss in the prior 12 months. These individuals consume

TABLE 1 | Descriptive statistics of consumption and expenditures.

Food group	Consumption (g/day)		Expenditure (\$/day)	
	Did not attempt weight loss <i>n</i> = 10,305	Attempted weight loss <i>n</i> = 6360	Did not attempt weight loss <i>n</i> = 10,305	Attempted weight loss <i>n</i> = 6360
Dry beans, peas, other legumes, nuts, and seeds				
Legumes	26.10 (90.40)	23.90 (79.00)	0.11 (0.61)	0.10 (0.39)
Nuts, nut butters, nut mixtures	10.10 (41.00)	11.70 (47.10)*	0.10 (0.39)	0.13 (0.45)*
Eggs				
Eggs	8.87 (28.20)	8.79 (27.80)	0.04 (0.16)	0.04 (0.17)
Egg mixtures	19.00 (57.10)	19.80 (58.30)	0.10 (0.38)	0.10 (0.40)
Fats, oils, and salad dressings				
Fats	2.53 (8.44)	2.14 (7.38)*	0.02 (0.07)	0.02 (0.07)*
Salad dressings	6.95 (21.00)	8.12 (21.50)*	0.04 (0.12)	0.04 (0.13)*
Fruits				
Citrus fruits, juices	47.50 (144.00)	44.00 (128.00)	0.10 (0.29)	0.10 (0.28)
Fruit juices (excluding citrus)	30.40 (131.00)	27.50 (117.00)	0.06 (0.31)	0.06 (0.32)
Other fruits	73.10 (132.00)	83.10 (152.00)*	0.30 (0.59)	0.36 (0.71)*
Grain products				
Cakes, cookies, pies, others	35.80 (73.20)	32.40 (69.90)*	0.31 (0.67)	0.29 (0.66)
Cereals (not cooked)	10.10 (27.20)	8.41 (23.00)*	0.06 (0.18)	0.05 (0.15)*
Crackers, snack products	12.00 (29.30)	12.30 (26.80)	0.11 (0.26)	0.11 (0.26)
Grain mixtures, frozen meals, soups	171.00 (267.00)	162.00 (246.00)*	0.96 (1.88)	0.95 (1.80)
Pancakes, waffles, others	5.90 (33.50)	4.96 (25.50)*	0.03 (0.18)	0.03 (0.15)
Pastas, rice, cooked cereals	61.10 (136.00)	52.10 (117.00)*	0.10 (0.53)	0.10 (0.40)
Quick breads	15.00 (47.40)	14.00 (43.30)	0.08 (0.30)	0.08 (0.29)
Yeast breads, rolls	46.00 (62.20)	41.60 (57.50)*	0.25 (0.38)	0.23 (0.37)*
Milk and milk products				
Cheeses	14.80 (35.50)	15.20 (35.70)	0.14 (0.33)	0.15 (0.34)
Creams, substitutes	7.73 (25.40)	8.86 (26.90)*	0.03 (0.12)	0.04 (0.12)*
Milk desserts, sauces	21.20 (64.30)	17.00 (56.10)*	0.09 (0.33)	0.08 (0.30)*
Milks, milk drinks, yogurts	147.00 (263.00)	125.00 (217.00)*	0.25 (0.59)	0.25 (0.59)
Meat, poultry, fish, and mixtures				
Beef	18.70 (60.00)	16.20 (50.20)*	0.30 (1.11)	0.27 (0.95)
Fish, shellfish	18.70 (75.50)	18.30 (72.10)	0.25 (1.43)	0.26 (1.74)
Frozen meals, soups	25.00 (118.00)	22.20 (106.00)	0.10 (0.55)	0.09 (0.51)
Meat, poultry, fish mixtures	96.10 (177.00)	86.80 (160.00)*	0.73 (1.62)	0.70 (1.52)
Frankfurters, sausages, lunchmeats	16.70 (43.20)	16.50 (42.30)	0.18 (0.50)	0.17 (0.49)

(Continues)

TABLE 1 | (Continued)

Meat, poultry, fish, and mixtures				
Pork	11.50 (39.80)	8.79 (33.90)*	0.15 (0.52)	0.12 (0.45)*
Poultry	37.80 (85.10)	37.00 (78.40)	0.36 (1.02)	0.36 (0.88)
Sugars, sweets, and beverages				
Soft drinks (non-diet)	223.00 (442.00)	153.00 (359.00)*	0.20 (0.39)	0.14 (0.32)*
Sugars, sweets	19.00 (44.00)	16.80 (38.50)*	0.13 (0.32)	0.12 (0.29)
Vegetables				
Dark-green vegetables	11.90 (49.10)	14.30 (56.60)*	0.06 (0.28)	0.08 (0.34)*
Orange vegetables	7.70 (37.30)	8.42 (39.80)	0.02 (0.13)	0.03 (0.15)
Other vegetables	62.80 (124.00)	67.40 (136.00)*	0.29 (0.68)	0.33 (0.70)*
Tomatoes, mixtures	23.70 (63.70)	27.00 (71.00)*	0.10 (0.25)	0.12 (0.30)*
Potatoes, starchy vegetables	49.40 (97.70)	44.40 (87.90)*	0.20 (0.40)	0.19 (0.37)*

Note: Values in parentheses are standard deviations. Asterisks (*) indicate estimates that are statistically different at the five percent level or lower between the two groups.

“Other fruits” (e.g., non-citrus fruits like apples) and most vegetables (except for potatoes and starchy vegetables) at a relatively higher volume than individuals who did not attempt weight loss. Conversely, they consume most grain products, sugars and sweets, soft drinks, milk and yogurt, and milk-based desserts at relatively lower volumes. Additionally, beef, pork, and meat-based food mixtures (i.e., combination food items with meat as the primary ingredient) are consumed at comparatively lower volumes by the intentional weight loss group, but other meats are consumed at around the same volume. These differences are visualized further in Figure 1 as “percentage changes,” where we take the average consumption of those who attempted weight loss in the prior 12 months less those who did not, divide by the latter, and multiply by 100. Last, expenditure differences between the two groups are similar to consumption differences in terms of direction. However, these are not always statistically different at the five percent level. This suggests that food expenditures may not move in the same direction as consumption when weight loss is attempted if the per-unit monetary value of foods purchased also changes. Alternatively, the typically higher annual household incomes among individuals who attempt weight loss (see Appendix Table A2) may be driving discrepancies between consumption and expenditures, reflecting a clear endogeneity issue that we address in our empirical strategy.

It should be noted that nonconsumption and associated zero expenditures occur for certain food groups and individuals. Appendix Table A3 depicts nonconsumption rates by weight loss history. Briefly, individuals who attempted weight loss in the prior 12 months report nonconsumption of “other vegetables” (e.g., cucumbers, lettuce, onions) at a rate 3.7 percentage points lower than individuals who did not attempt weight loss. Conversely, they report nonconsumption of soft drinks at a rate 7.5 percentage points higher. The nonconsumption rates between consumer groups are within 3 percentage points of each other for all other food groups. Additionally, the highest rate of nonconsumption for both consumer groups occurs for “Pancakes, waffles, others” at 95

percent of individuals. Though this is a large share of our sample, there are still over 800 individuals reporting some positive level of consumption in this most extreme case. However, the censored nature of the NHANES dietary recall data requires consideration in our empirical approach.

3.2 | Conceptual Model

Food consumption can be described in reduced form as:

$$C_{ji} = \alpha_j^c + \gamma_j^c Treatment_i + \sum_{k=1}^K \beta_{jk}^c X_i + \varepsilon_{ji}^c \quad \forall j = 1 \dots 35 \quad (1)$$

where j indexes food groups; i indexes individuals; $Treatment_i$ is a treatment indicator that is equal to one when an individual reports having attempted to lose weight in the prior 12 months and is equal to zero otherwise; X_i is a vector of controls for consumer characteristics; ε_{ji}^c is a random error term; and α_j^c , γ_j^c , and β_{jk}^c are parameters to be estimated. Food expenditures M_{ji} are described similarly by:

$$M_{ji} = \alpha_j^m + \gamma_j^m Treatment_i + \sum_{k=1}^K \beta_{jk}^m X_i + \varepsilon_{ji}^m \quad \forall j = 1 \dots 35. \quad (2)$$

Importantly, there is no mechanism by which prior-day food consumption and expenditures influence a decision made to lose weight in the prior 12 months (unless, of course, that decision happened to be made during the prior day). As such, there is no concern over potential simultaneity in these proposed reduced form models.

3.3 | Reducing Confounding

Attempting to lose weight is an endogenous choice made by a group of individuals that are already likely to exhibit different

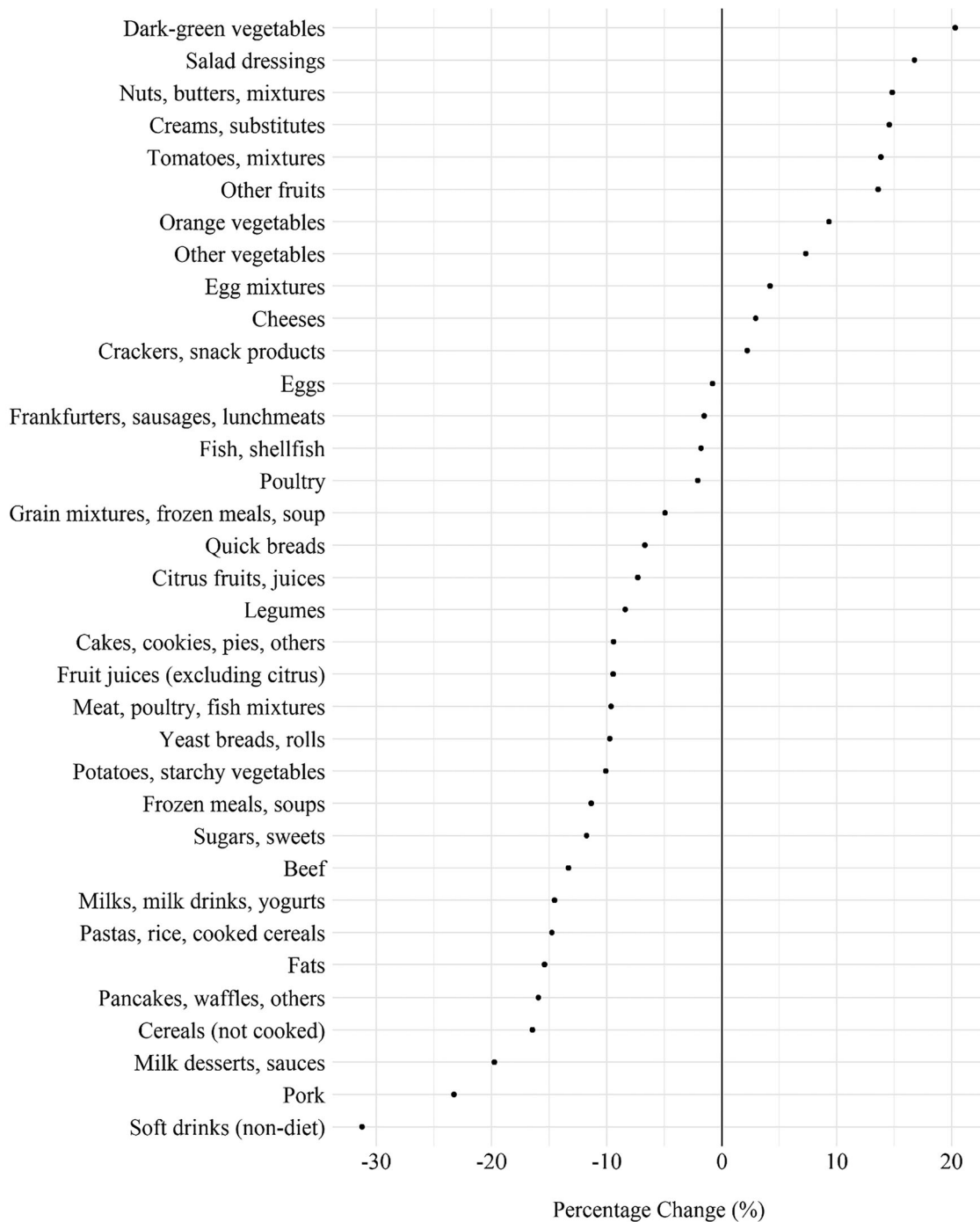


FIGURE 1 | Food consumption differences—did not attempt weight loss versus attempted weight loss. Percentage changes are calculated by taking the average consumption of individuals who attempted weight loss in the prior 12 months *less* the consumption of individuals who did not, dividing by the latter, and multiplying by 100.

food purchasing behavior. From reporting of NHANES participants' characteristics by weight loss history (Appendix Table A2), certain sociodemographic groups more frequently attempt to lose weight than others (i.e., females, younger individuals, non-Hispanic white individuals, higher earners, etc.). Prior work using similar data notes that US consumption patterns vary by sex, race, and other factors (Wang et al. 2010; Zeng et al. 2019). As such, estimates of the impact of weight loss attempts on consumer behavior may be driven in part by differences in consumers characteristics rather than by intentional weight loss itself.

Accordingly, we take a selection on observables approach to account for these confounding influences of consumer characteristics. Selection on observables assumes that, after controlling for all observable confounders (e.g., sex, income, etc.), the decision to attempt weight loss is as good as randomly assigned. Formally, this is accomplished by using coarsened exact matching [CEM] (Iacus et al. 2011) to balance characteristics between individuals who attempted weight loss in the prior 12 months and those who did not, which eliminates the confounding influence of observed, secondary consumer characteristics that

drive both the choice to lose weight and choices made in food markets.

Exact matching, which is a more stringent version of CEM, serves to replicate a randomized experiment where treated and control subgroups are randomly different on all covariates (Cochran 1968; Iacus et al. 2011; Stuart 2010). However, disadvantages of exact matching include the loss of many observations when data are high dimensional and matches cannot be made and, commonly, the inability of researchers to observe all potentially confounding characteristics of consumers. Although no matching method can alleviate the latter limitation, CEM reduces the loss of observations by simply matching on a broader range of measured covariates (e.g., matching on categorical rather than continuous income). The procedure entails placing treated and control units into strata such that all units in a stratum are perfectly identical across all measured, but coarsened, covariates (Iacus et al. 2011; Vass et al. 2022). Thus, characteristics are perfectly balanced both within a stratum and on average between the treated and control subgroups. No assumptions on functional form are required to eliminate confounding due to these measured covariates. We prefer this approach to others, such as propensity score matching, that require the proper estimation of a first-stage distance measure and often other adjustments in the matching process to avoid poor matches (Stuart 2010).

In our use of CEM, the treatment variable is an NHANES participant having attempted to lose weight in the prior 12 months, while the matching variables are participants' survey cycle, sex, age, race, educational attainment, and annual household income. Participants' body mass index (BMI) category and physical exercise behavior are not included in the matching process as these characteristics are necessarily (or likely) impacted by the treatment, creating selection bias comparable in size to bias from classical confounding if included (Greenland 2003; Stuart 2010).

After matching (i.e., after stratum assignment), stratum propensity scores (SP) are calculated as the proportion of participants in each stratum that are treated. All participants in a stratum then receive the same SP. Treated participants are assigned a matching weight of one (i.e., they are left unweighted), while control participants are assigned a weight of $SP/(1-SP)$. This ensures that the weighted average of measured covariates are balanced between each subgroup (Austin and Stuart 2015; Stuart 2010; Vass et al. 2022). Matching weights are then multiplied by NHANES dietary day one sample weights, adjusted according to NCHS (National Center for Health Statistics 2024b) suggested methods to account for the combination of four NHANES survey cycles (2011–2012 to 2017–2018). This final set of matching weights, normalized to mean of one, provides generalizable estimates and is used in the outcome and effect estimations discussed in the ensuing subsections. Matching and weight calculations are conducted using the MatchIt package (Ho et al. 2024) in R version 4.4.1.

3.4 | Outcome Estimation

We estimate consumption and expenditure outcomes using raw dietary intakes (i.e., NHANES-reported intakes), rather than *usual* dietary intakes (i.e., the long-term average daily intake of

an individual). For episodically consumed foods like “Pancakes, waffles, others”, using raw dietary intakes does not consider measurement error inherent in the food consumption recall and can reflect an overestimate of the true number of non-consumers (Bina and Tonsor 2024). Various research efforts using NHANES dietary data implement the National Cancer Institute (NCI) method to calculate usual intakes (Morris and Jacques 2013; O'Connor et al. 2022; Pasiakos et al. 2015). Following Bina and Tonsor (2024), we elect to use raw dietary intakes in our analysis as the NCI method does not allow for nonconsumption (i.e., it does not allow for predicted usual intakes of zero)—which may be “true” for some individuals and food groups—and because mismeasured dietary intakes (i.e., the outcome variable) will not bias parameter estimates if the measurement error is random and independent of the regressors (Wooldridge 2010, pp. 76–78).

To correct for censored values, food consumption and expenditures described by Equations (1) and (2) are formally estimated using a censored regression approach where the log-likelihood function of the lower-limit censored model (censored at zero grams or zero dollars) is:

$$l_j = \sum_{i \in (y_{ji} > 0)} \ln \left[\phi \left(\frac{y_{ji} - X_i' \beta}{\sigma} \right) / \sigma \right] + \sum_{i \in (y_{ji} = 0)} \ln \left[\Phi \left(\frac{0 - X_i' \beta}{\sigma} \right) \right] \quad \forall j = 1 \dots 35 \quad (3)$$

and where y_{ji} is individual i 's consumption of or expenditures on the 35 evaluated food groups. The vector of controls X_i includes the participant characteristics used in the matching process (denoted in Table A2) along with the treatment indicator ($Treatment_i$), BMI category, and physical exercise behavior.¹ This specification reflects a Type I Tobit model (Tobin 1958). Equation (3) is estimated separately for each food group j , outcome of interest (i.e., consumption and expenditure), and sample (i.e., full sample, matched samples) via maximum likelihood and using the QLIM procedure in SAS 9.4. Weights obtained from the CEM procedure are incorporated into subsequent outcome estimations, but the full sample outcome estimations are left unweighted (thus reflecting associations and not treatment effects).

Expected values and marginal effects for the Type I Tobit model are described succinctly by McDonald and Moffitt (1980) and Wooldridge (2010, pp. 671–676). We focus on the unconditional expected values of the dependent variable (censored and uncensored), which reduce to:

$$E(y|X) = \Phi \left(\frac{X\beta}{\sigma} \right) X\beta + \sigma \phi \left(\frac{X\beta}{\sigma} \right) \quad (4)$$

where the marginal effect of a discrete regressor is simply the difference in $E(y|X)$ when the regressor is equal to zero and when it is equal to one.

3.5 | Effect Estimation

Following the weighted estimation of Equation (3), we implement a doubly robust standardization procedure (Chatton and Rohrer

2024; Funk et al. 2011; Snowden et al. 2011; Vansteelandt and Keiding 2011). First, we predict unconditional expected values for every participant in the treated sample assuming that they are treated (i.e., $Treatment_i = 1$). Second, we predict unconditional expected values for every participant in the treated sample assuming that they are untreated (i.e., $Treatment_i = 0$). Third, we compute the average of each estimated potential outcome across the entire treated sample, resulting in two average estimated potential outcomes. Finally, we take the difference between the averages.

In causal inference efforts, this procedure yields a doubly robust estimate of the average treatment effect on the treated (ATT) if either the matching procedure sufficiently reduces imbalance in the covariates between control and treated groups or the outcome model is correctly specified (Chatton and Rohrer 2024; Funk et al. 2011). In our application, we recognize the limitation of the selection on observables approach and the possibility of unmeasured confounders. Thus, the effects of weight loss attempts presented in this work may not reflect “causal” impacts on consumer behavior in US food markets. Rather, these effects account for some, but not all, of the variation in characteristics between consumer groups and are compared to simple associations (i.e., traditionally-estimated unconditional marginal effects at the means).

Krinsky and Robb (1986) confidence intervals are constructed to assess certainty in the outcome and effect estimations. Briefly, this involves resampling from the original censored regression parameter estimates using a multivariate normal distribution parameterized by the original estimates (i.e., the means) and covariance matrix (i.e., the variance). For each draw, marginal effects and treatment effects are calculated, thus creating an empirical distribution for each measure of interest. We use 1000 draws in our resampling procedure.

3.6 | Heterogeneity Assessments

It should be considered that consumers may display fundamentally different consumption patterns between food outlets. Todd et al. (2010) note that food-away-from-home (FAFH) dining increases caloric intake and reduces diet quality. In our application, this may meaningfully alter how our sample makes dietary choices. Accordingly, we utilize NHANES reporting of food sources to determine how intentional weight loss affects consumption and expenditures in both food at home (FAH) and FAFH. Specifically, we define FAH as food sourced from a grocery store, supermarket, or caught/grown at home and FAFH as food sourced from a restaurant, bar, vending machine, sporting event, street vendor, or convenience store (National Center for Health Statistics 2024a). All analyses discussed above are then conducted separately for FAH and FAFH dietary reporting.

Additionally, we are unable to determine if NHANES participants were actively attempting to lose weight during their dietary recall period. Thus, estimated treatment effects will encompass both short-run dietary changes resulting from current weight loss attempts and/or longer-run dietary changes resulting from weight loss attempts occurring any time in the prior 12 months, but not necessarily ongoing. Though we cannot truly ascertain

short- versus long-run impacts of intentional weight loss, we provide a comparison between subgroups of consumers that likely have different prevalence of attempting weight loss *during* their MEC exam. Specifically, we distinguish between NHANES participants who are categorized as under- or normal weight [BMI less than 25.0 kg/m²] and those that are categorized as overweight or obese [BMI greater than or equal to 25.0 kg/m²] (Centers for Disease Control and Prevention 2024a). We then conduct all aforementioned statistical analyses (i.e., CEM, outcome estimation, effect estimation) separately across the two groups. Our argument here is that participants who attempted weight loss and are still overweight or obese are more likely to still be in a weight loss stage during their dietary recall *relative to* participants who attempted weight loss and are under- or normal weight (thus suggesting a successful, completed weight loss period).

4 | Results

Interpretation of parameter estimates of covariates in the outcome model is often done misleadingly, creating confusion between direct effects with total effects (Westreich and Greenland 2013). In addition, our chosen outcome models are lower-limit censored models (i.e., Type I Tobit) and the parameter estimates represent marginal effects on the latent outcomes. These are of little value. As such, we report and discuss only (i) “traditional” marginal effects of intentional weight loss attempts on unconditional expected values and (ii) treatment effects of intentional weight loss attempts on unconditional expected values (i.e., effects derived using matching and doubly robust standardization).

4.1 | Consumption Effect Estimates

Table 2 depicts the marginal effects and treatment effects of prior-year weight loss attempts on the unconditional expected values of consumption for each of the 35 evaluated food groups. The effects on conditional expected values are similarly reported in Appendix Table A4.

From column 3 (“Full”), prior-year weight loss attempts have a positive and statistically significant effect on daily consumption of nuts and nut butters; eggs; egg mixtures (i.e., mixture products whose primary ingredient is eggs; e.g., egg omelet); salad dressings; “other fruits” (i.e., non-citrus fruit; e.g., bananas, apples, strawberries); creams and cream substitutes; frankfurters, sausages, and lunchmeat; poultry; dark-green vegetables; orange vegetables; “other vegetables” (e.g., lettuce, onions, cucumbers); and tomatoes. Though these treatment effects appear small in magnitude from Table 2 reporting, they range from 9.8 percent (egg mixtures) to 37.8 percent (nuts and nut butters) over the average consumption of individuals who did not attempt weight loss in the prior 12 months. Importantly, most of these food groups are “single-ingredient” or minimally processed. The notable, and surprising, example is frankfurters, sausages, and lunchmeat, which are generally not viewed as healthful foods. We have no definitive explanation for this result but do note that the nutrition literature has found high-protein diets to provide several potential benefits during weight loss, including increased feelings of satiety and elevated muscle mass retention (Halton and Hu 2004; Leidy et al. 2015; Paddon-Jones et al. 2008), which may make these

TABLE 2 | Effects of prior-year weight loss attempts on unconditional expected food consumption.

Food group	Marginal effects (g/day)	Average treatment effect on the treated (g/day)				
		Full ^a	FAH	FAFH	BMI < 25.0	BMI ≥ 25.0
Dry beans, peas, other legumes, nuts, and seeds						
Legumes	0.82	1.35	1.89	−0.06	3.05	1.37
Nuts, nut butters, nut mixtures	3.02*	3.82*	3.68*	0.27	8.24*	2.66*
Eggs						
Eggs	0.48	0.98*	0.89*	−0.25	−0.67	0.89
Egg mixtures	1.18	1.87*	0.61	2.27*	3.67	2.16*
Fats, oils, and salad dressings						
Fats	−0.17	−0.23*	−0.18	−0.03	0.81*	−0.48*
Salad dressings	0.84*	1.26*	0.57*	0.86*	2.94*	0.70
Fruits						
Citrus fruits, juices	−2.33	−2.77	−3.23	0.38	5.03	−5.09*
Fruit juices (excluding citrus)	−3.52	−2.32	−1.29	−1.31	2.61	−5.57*
Other fruits	7.41*	9.40*	6.50*	2.39*	16.64*	8.41*
Grain products						
Cakes, cookies, pies, others	−2.73*	−3.18*	−2.70*	0.57	−9.28*	−3.00*
Cereals (not cooked)	−1.51*	−1.94*	−2.07*	0.03	−3.07*	−1.89*
Crackers, snack products	0.31	0.37	0.48	−0.65	1.32	0.13
Grain mixtures, frozen meals, soup	−8.02*	−10.97*	−7.12*	−8.29*	−31.55*	−6.45
Pancakes, waffles, others	−1.47*	−2.03*	−1.64*	0.03	−2.44*	−2.22*
Pastas, rice, cooked cereals	−0.99	−1.17	−1.06	1.28	−6.03	−0.10
Quick breads	−0.41	−0.02	−0.27	0.05	1.25	−0.70
Yeast breads, rolls	−2.15*	−2.49*	−2.83*	0.30	−3.53	−2.30
Milk and milk products						
Cheeses	−1.57*	−1.90*	−1.91*	−0.55	−0.56	−2.53*
Creams, substitutes	0.94*	0.94*	0.10	0.72*	0.44	1.10*
Milk desserts, sauces	−2.74*	−3.32*	−4.48*	1.74*	−5.64*	−2.36*
Milks, milk drinks, yogurts	−11.46*	−11.95*	−9.32*	−2.50	−29.92*	−6.49
Meat, poultry, fish, and mixtures						
Beef	−0.55	−0.99	−0.15	−1.18	−2.08	−1.28
Fish, shellfish	−0.75	−0.71	0.92	−2.05*	−0.96	−0.79
Frozen meals, soups	2.03	2.26	0.22	1.99*	12.75*	−0.73
Meat, poultry, fish mixtures	−5.56*	−5.22*	−2.68	1.07	−5.77	−5.19
Frankfurters, sausages, lunchmeats	2.05*	2.29*	1.42*	0.81	0.59	2.96*
Pork	−0.23	−0.13	0.41	−0.72*	0.47	−0.68
Poultry	2.63	4.13*	1.44	3.26*	9.91*	2.50

(Continues)

TABLE 2 | (Continued)

Sugars, sweets, and beverages						
Soft drinks (non-diet)	−60.43*	−62.20*	−40.57*	−34.54*	−29.24*	−74.29*
Sugars, sweets	−0.68	−0.61	−0.15	−0.51	0.83	−1.05
Vegetables						
Dark-green vegetables	3.90*	3.95*	3.44*	0.71	6.73*	3.55*
Orange vegetables	0.94*	1.29*	0.17	0.57*	0.93	1.32*
Other vegetables	8.15*	9.66*	9.06*	0.42	20.19*	6.45*
Tomatoes, mixtures	2.64*	3.72*	2.25*	2.31*	7.65*	2.57*
Potatoes, starchy vegetables	−3.99*	−5.40*	−3.75*	−1.99	−7.22*	−4.64*

^aData are: Full = the matched sample obtained from applying coarsened exact matching to the full usable sample; FAH = the usable sample is restricted to FAH consumption prior to matching; FAFH = the usable sample is restricted to FAFH consumption prior to matching; BMI < 25.0 = the usable sample is restricted to individuals whose BMI is less than 25.0 kg/m² prior to matching; BMI ≥ 25.0 = the usable sample is restricted to individuals whose BMI is greater than or equal to 25.0 kg/m² prior to matching. Asterisks (*) denote estimates that are statistically significant at the five percent level or lower.

products more attractive to individuals who are attempting to lose weight.

To place these results in a national context, the average US population from 2011 to 2018 was 319,459,812 people (Federal Reserve Bank of St. Louis 2024). Using NHANES-reported weight loss history and weighting responses by dietary day one sample weights (National Center for Health Statistics 2024b), 129,357,296 US citizens (roughly 40.5 percent of the entire population) attempted to lose weight each year, on average. Extrapolating to the national level, our estimates indicate that intentional weight loss attempts yield increases in daily, national consumption of: nuts and nut butters of 1.1 million pounds (494 million grams); eggs of 0.3 million pounds; other fruits of 2.7 million pounds; frankfurters, sausages, and lunchmeats of 0.7 million pounds; poultry of 1.2 million pounds; and vegetables and tomatoes of 5.3 million pounds.

Column 3 of Table 2 (“Full”) further shows that prior-year weight loss attempts have negative and statistically significant effects on daily consumption of fats (e.g., butter, margarine); cakes, cookies, pies, and other baked goods; cereals (i.e., breakfast cereals); grain mixtures, frozen meals, and soup (i.e., mixture products whose primary ingredient is grain; e.g., macaroni with cheese, spaghetti with tomato sauce); pancakes, waffles, and other grain-based breakfast products; yeast breads and rolls; cheeses; milk-based desserts (e.g., ice cream); milk and yogurt; meat, poultry, and fish mixtures (i.e., mixture products whose primary ingredient is meat; e.g., meat loaf, tuna salad); soft drinks; and potatoes. These effects reflect reductions in consumption ranging from 5.4 percent (yeast breads and rolls) to 34.4 percent (pancakes, waffles, and other grain-based breakfast products) below the average levels reported by individuals who did not attempt weight loss in the prior 12 months. Further, and in contrast to the positive treatment effects discussed above, negative effects are generally observed for products that are highly processed, such as baked goods and soft drinks. However, dairy products and carbohydrate-dense potatoes are also impacted by weight loss efforts. Given the processed nature of many of these products, it is difficult to quantify volume impacts at a national level. We note, though, that these results reflect a reduction in daily, national consumption

of potatoes and starchy vegetables of 1.5 million pounds. Further, the −62.2 gram per day (g/day) impact on soft drink consumption represents a 27.9 percent reduction from the average consumption of individuals who did not attempt to lose weight.

Now considering potential consumption differences between FAH and FAFH (columns 4 and 5 of Table 2), treatment effects are positive and statistically significant in both outlets for salad dressings, other fruits, and tomatoes. Conversely, they are negative and statistically significant in both outlets for grain mixtures, frozen meals, and soup; and soft drinks. Like the primary results, FAH consumption of many grain products is negatively impacted by prior-year weight loss attempts; however, FAFH consumption is generally not. These disparities between outlets extend to vegetable products, with weight loss attempts having positive impacts on FAH dark-green and other vegetable consumption and negative impacts on FAH potato consumption, while having positive impacts on FAFH orange vegetable consumption. Meanwhile, in the broader meat category, treatment effects are positive for FAH frankfurters, sausages, and lunchmeats; are positive for FAFH “frozen meals, soups” (i.e., those with meat as the primary ingredient) and poultry; and are negative for FAFH fish and pork. Further differences are observed in the milk product category with negative effects observed for FAH cheeses and milk and yogurt, and positive effects observed for FAFH creams and cream substitutes. Interestingly, treatment effects on milk-based desserts have opposite signs between outlets (negative for FAH and positive for FAFH) that are both statistically significant. In all, these results highlight that consumers’ decisions to attempt weight loss affect food groups differently due to either (i) broader changes in where food is being consumed, (ii) perceptions on the differences in how foods are prepared between outlets and their associated caloric content (e.g., oven-baked French fries at home vs. deep-fried French fries away from home), or (iii) simple tendencies to consume certain foods only in a specific outlet (e.g., consuming breakfast cereals only at home).

Across both BMI assessments (columns 6 and 7 of Table 2), prior-year weight loss attempts have positive and statistically significant effects on the consumption of nuts and nut butters; other fruits; dark-green vegetables; other vegetables; and tomatoes, which is

all consistent with the primary results. Further, negative effects are observed in each BMI-based assessment for cakes, cookies, pies, and other baked goods; cereals; pancakes, waffles, and other grain-based breakfast foods; milk-based desserts; soft drinks; and potatoes. Again, these findings are consistent with the primary results. Interestingly, effect estimates are larger in magnitude in the lower-BMI assessment (i.e., BMI < 25.0 kg/m²) for nuts and nut butters; other fruits; each impacted grain product; milk-based desserts; and each impacted vegetable product. This may indicate a longer-term proclivity to shift consumption out of carbohydrate-dense desserts (milk- and grain-based) and grain-based breakfast foods and into single-ingredient nuts and non-starchy vegetables following a weight loss attempt. Conversely, negative effect estimates for soft drinks are much larger in magnitude in the heterogeneity assessment utilizing only higher-BMI individuals [i.e., BMI ≥ 25.0 kg/m²] (−29.2 g/day for the lower-BMI assessment vs. −74.3 g/day for the higher-BMI assessment). This may indicate that individuals who still have weight to lose—that is, they are still classified as overweight or obese (Centers for Disease Control and Prevention 2024a)—decrease their consumption of sugary soft drinks to a relatively higher degree. The consumption of fats, fruit juices, and cheeses may be similarly impacted by potentially more recent or current weight loss attempts, as indicated by negative and significant effect estimates in only the higher-BMI assessment. However, we reiterate that we cannot observe the true diet status and duration of NHANES participants who report having attempted weight loss in the prior 12 months.

4.2 | Expenditure Effect Estimates

Table 3 depicts the marginal effects and treatment effects of prior-year weight loss attempts on the unconditional expected values of expenditures for each of the 35 evaluated food groups. The effects on conditional expected values are similarly reported in Appendix Table A5.

Prior-year weight loss attempts have positive and statistically significant effects (column 3 of Table 3) on daily expenditures on nuts and nut butters (4.17 cents per day [¢/day]); salad dressings (0.73 ¢/day); other fruits (4.17 ¢/day); creams and cream substitutes (0.40 ¢/day); frankfurters, sausages, and lunchmeats (2.25 ¢/day); poultry (2.98 ¢/day); dark-green vegetables (2.50 ¢/day); orange vegetables (0.48 ¢/day); other vegetables (4.66 ¢/day); and tomatoes (1.98 ¢/day). Again pairing US population estimates for the 2011–2018 period (Federal Reserve Bank of St. Louis 2024) with NHANES-reported weight loss history, these estimates reflect increases in daily, national expenditures of \$5.4 million (nuts and nut butters); \$0.9 million (salad dressings); \$5.4 million (other fruit); \$0.5 million (creams and cream substitutes); \$2.9 million (frankfurters, sausages, and lunchmeats); \$3.9 million (poultry); \$3.2 million (dark-green vegetables); \$0.6 million (orange vegetables); \$6.0 million (other vegetables); and \$2.6 million (tomatoes).

Conversely, the effects of prior-year weight loss attempts on food expenditures (column 3 of Table 3) are negative and statistically significant for cakes, cookies, pies, and other baked goods (−2.67 ¢/day); cereals (−1.39 ¢/day); grain mixtures, frozen meals, and soups (−7.99 ¢/day); pancakes, waffles, and other grain-based breakfast products (−1.08 ¢/day); yeast breads and rolls (−1.38

¢/day); cheeses (−1.21 ¢/day); milk-based desserts (−1.62 ¢/day); soft drinks (−5.61 ¢/day); and potatoes (−1.61 ¢/day). The daily, national reductions in expenditures are, correspondingly: \$3.5 million (cakes, cookies, pies, and other baked goods), \$1.8 million (cereals), \$10.3 million (grain mixtures, frozen meals, and soups), \$1.4 million (pancakes, waffles, and other grain-based breakfast foods), \$1.8 million (yeast breads and rolls), \$1.6 million (cheeses), \$2.1 million (milk-based desserts), \$7.3 million (soft drinks), and \$2.1 million (potatoes).

Food expenditure results across food outlet and BMI heterogeneity assessments are broadly consistent with the consumption results discussed previously. For brevity, we forgo further reporting of those estimates.

4.3 | Considerations

Several important considerations need to be made when interpreting our estimates of the effects of weight loss attempts on food-related consumer behavior. First is the period of study. The NHANES project was disrupted during the coronavirus disease 2019 pandemic and, as such, the last full 2-year cycle was for 2017–2018. Data for an August 2021–2023 survey cycle is being made available gradually. However, at time of writing, important information related to participants' incomes, educational attainment, and weight loss attempts was not yet available. Further, the Purchase to Plate prices necessary for estimation of August 2021–2023 food expenditures will not be provided for several years (US Department of Agriculture Economic Research Service 2024). Since we implemented a matching procedure that reduced any confounding effects of time (i.e., survey cycle), we do not believe our estimates would be influenced by future price updates. However, it must also be considered that the recent adoption of GLP-1 medications may alter food consumption patterns in a way that is different than the “traditional” weight loss strategies reflected in this study. That is, the hormonal changes induced by these medications—making it harder to “cheat” on a diet—could yield GLP-1 treatment effects that differ from our results. However, only three percent of US adults indicated using the medications to lose weight in 2024 (Witters and Maese 2024), compared to a substantially larger 27 percent that reported seriously attempting weight loss using any strategy (Brenan 2024). Thus, we believe our findings are representative of the majority share of US consumers that are attempting to lose weight through methods other than weight loss medications.

Second, we do not know for what length of time an NHANES participant attempted to lose weight or the intensity of that attempt (i.e., how much weight they were trying to lose). More important, we do not observe if participants are actively trying to lose weight during their dietary recall. These limitations, in addition to the cross-sectional nature of the NHANES project, preclude us from distinguishing between short- and long-run effects of weight loss attempts on consumer behavior or quantifying any nonlinear effects of weight loss intensity. We do note, however, that since some portion of participants may not have been actively losing weight during their dietary recall, our estimates of the effects of weight loss attempts on food consumption and expenditures are likely understated. Further, our heterogeneity assessment across

TABLE 3 | Effects of prior-year weight loss attempts on unconditional expected food expenditures.

Food group	Marginal effects (¢/day)	Average treatment effect on the treated (¢/day)				
		Full ^a	FAH	FAFH	BMI < 25.0	BMI ≥ 25.0
Dry beans, peas, other legumes, nuts, and seeds						
Legumes	−0.12	0.16	0.48	0.02	1.61	0.04
Nuts, nut butters, nut mixtures	3.30*	4.17*	4.08*	0.21	6.91*	3.32*
Eggs						
Eggs	0.24	0.51	0.40*	−0.23	−0.31	0.45
Egg mixtures	0.28	0.66	0.29	1.39	1.10	1.04
Fats, oils, and salad dressings						
Fats	−0.15	−0.18	−0.10	−0.09	0.68*	−0.35*
Salad dressings	0.49*	0.73*	0.31*	0.53*	1.84*	0.39
Fruits						
Citrus fruits, juices	−0.41	−0.52	−0.61	0.06	1.27	−1.12*
Fruit juices (excluding citrus)	−0.64	−0.39	−0.17	−0.33	0.75	−1.04*
Other fruits	3.23*	4.17*	2.66*	1.19*	6.77*	3.88*
Grain products						
Cakes, cookies, pies, others	−2.09*	−2.67*	−2.13*	−0.01	−9.07*	−2.34*
Cereals (not cooked)	−1.08*	−1.39*	−1.50*	0.05	−2.97*	−1.16*
Crackers, snack products	0.27	0.34	0.33	−0.41	1.24	0.09
Grain mixtures, frozen meals, soup	−6.61*	−7.99*	−2.53	−8.79*	−21.24*	−5.04
Pancakes, waffles, others	−0.80*	−1.08*	−0.86*	0.02	−1.46*	−1.01*
Pastas, rice, cooked cereals	−0.31	−0.35	−0.28	0.44	−1.29	0.00
Quick breads	−0.13	0.08	−0.11	0.06	1.40	−0.52
Yeast breads, rolls	−1.20*	−1.38*	−1.56*	0.11	−2.07	−1.08
Milk and milk products						
Cheeses	−1.09*	−1.21*	−1.48*	−0.33	−0.34	−1.56*
Creams, substitutes	0.40*	0.40*	−0.03	0.41*	0.39	0.43*
Milk desserts, sauces	−1.27*	−1.62*	−2.11*	0.68	−2.23*	−1.24*
Milks, milk drinks, yogurts	−0.63	−0.51	0.02	−1.11	−6.18*	0.94
Meat, poultry, fish, and mixtures						
Beef	−1.37	−2.06	−0.40	−2.39	−3.53	−2.81
Fish, shellfish	−1.56	−1.32	1.31	−3.00	−4.16	−1.26
Frozen meals, soups	1.04	1.07	−0.08	1.00*	7.75*	−0.67
Meat, poultry, fish mixtures	−2.71	−1.48	−0.79	2.58	1.24	−1.40
Frankfurters, sausages, lunchmeats	2.04*	2.25*	1.32	0.86	0.82	2.95*
Pork	0.15	0.75	1.76*	−1.08*	1.33	0.33
Poultry	1.58	2.98*	1.17	2.32	7.15*	1.59

(Continues)

TABLE 3 | (Continued)

Sugars, sweets, and beverages						
Soft drinks (non-diet)	−5.43*	−5.61*	−3.65*	−3.12*	−2.57*	−6.74*
Sugars, sweets	−0.31	−0.16	0.22	−0.42	2.00*	−0.69
Vegetables						
Dark-green vegetables	2.24*	2.50*	2.22*	0.34	5.49*	1.87*
Orange vegetables	0.34*	0.48*	0.07	0.22*	0.12	0.53*
Other vegetables	3.73*	4.66*	4.52*	0.50	4.74	4.37*
Tomatoes, mixtures	1.44*	1.98*	1.10*	1.23*	3.69*	1.52*
Potatoes, starchy vegetables	−1.14*	−1.61*	−1.35*	−0.73	−2.09	−1.41

^aData are: Full = the matched sample obtained from applying coarsened exact matching to the full usable sample; FAH = the usable sample is restricted to FAH consumption prior to matching; FAFH = the usable sample is restricted to FAFH consumption prior to matching; BMI < 25.0 = the usable sample is restricted to individuals whose BMI is less than 25.0 kg/m² prior to matching; BMI ≥ 25.0 = the usable sample is restricted to individuals whose BMI is greater than or equal to 25.0 kg/m² prior to matching. Asterisks (*) denote estimates that are statistically significant at the five percent level or lower.

BMI status speaks indirectly to the issue of short- versus long-run effects, though this issue remains a limitation of the study.

Last, our aggregation of consumption to 35 food groups may mask transitions in consumption and expenditures *within* a group as consumers attempt to lose weight. That is, our results do not speak to substitution patterns within broad food groups and it is probable that weight loss attempts motivate consumers to eat and spend less (or more) on certain disaggregated products within a broader product category. An example of this may be consumers switching consumption from 80 percent lean ground beef to 90 percent lean ground beef. However, given the high degree of nonconsumption at the product level and an inability to map prices to specific food components (e.g., beef in a cheeseburger), our chosen level of aggregation was most appropriate for the research objective. Such aggregation issues will likely also plague future work in this area using other data sources.

5 | Discussion

Weight management is increasingly important in the US and its economic impacts in the food industry are not well understood. This research aimed to fill that knowledge gap by estimating the effects of weight loss attempts on consumption and expenditures across 35 food groups. Attempting weight loss yields increases in daily, national consumption of, primarily, single-ingredient food groups, including nuts and nut butters; non-citrus fruits; non-starchy vegetables and tomatoes; and eggs and poultry. Surprisingly, the consumption of frankfurters, sausages, and lunchmeats is also positively affected by weight loss efforts, which contrasts with polarizing views on the healthfulness of red meat and processed meat (Sievert et al. 2022) and observed detrimental health impacts (Farvid et al. 2021; Grosso et al. 2022). Conversely, prior-year weight loss attempts yield reductions in daily, national consumption of many highly processed and carbohydrate-dense foods like grain-based baked goods, breakfast cereals, and soft drinks. However, dairy products and potatoes are likewise negatively affected. The associated monetary impacts are sizable, ranging from decreases in daily, national expenditures

on soft drinks and processed grain products of \$7.3 million and \$18.8 million, respectively, to increases in daily, national expenditures on non-citrus fruits and non-starchy vegetables of \$17.8 million.

These results come with a host of policy and industry implications. First, government-led healthy eating campaigns (e.g., MyPlate, Supplemental Nutrition Assistance Program [SNAP] Education) may have different impacts on various sectors of the US food industry as suggested by the disparate effects of weight loss across grain-based food product and produce expenditures. This is not particularly surprising as these campaigns commonly advocate for the consumption of certain food groups. For example, USDA's MyPlate suggests that half of an individual's plate of food should be composed of fruits and vegetables (US Department of Agriculture 2025). As such, the potential, disparate economic impacts of these efforts on food manufacturers, retailers, and commodity producers should be more seriously considered. Specifically, those involved in public health and nutrition efforts should be cognizant that, if successful in altering consumers' food intake patterns, those efforts are holistic in nature. That is, initiatives to improve dietary quality may affect not only health outcomes, but the economic wellbeing of those involved in food production.

Tangentially related, in-kind benefits provided through USDA Food and Nutrition Services' SNAP program may be utilized differently as recipients attempt weight loss. Though the summary statistics we present in Appendix Table A2 indicate that higher-earning individuals more frequently attempt to lose weight, a sizable portion of our sample both attempted to lose weight in the prior year and earned less than \$20,000 annually (roughly 7 percent). Intuitively, household monthly SNAP allotments (US Department of Agriculture Food and Nutrition Service 2025) may become less restrictive as individuals in the household attempt weight loss and, correspondingly, reduce total food consumption. However, it is possible that changes in the diet entail increases in expenditures per unit of food (e.g., the observed shift in expenditures from processed grain products to vegetables). Thus, allotments may become increasingly restrictive as low-income consumers make health-focused dietary changes. This

is a distinctly different concept than the relationships between SNAP and food security or obesity that have been assessed in the past (Almada et al. 2016; Gundersen et al. 2017) and should be considered when crafting policy in the intersection of food spending, healthy eating, and income.

Last, different economic outcomes will be experienced by different sectors of the US food industry as society's focus on weight management increases. The growing prevalence of GLP-1 medications and government decisions to cover GLP-1 use under Medicare Part D (Aboulenein and Wingrove 2024) may further exacerbate these disparate outcomes as new users change their eating behavior. Although the economic impacts of weight loss are most directly observed by food manufacturers and retailers, changes in the derived demand for food inputs may correspondingly affect commodity producers. As an example, food grains (i.e., rice, rye, and wheat) account for a meaningful portion of domestic agricultural production with cash receipts totaling \$15.6 billion in 2023 (US Department of Agriculture Economic Research Service 2025). Reductions in food consumption resulting from weight loss attempts may be disproportionately borne by this sector—as evident by our findings related to processed, grain-based food consumption—with lower derived demand resulting in lower farmgate prices. However, given the complexity of the US food system, farm-level impacts will likely also depend on farm-to-wholesale and wholesale-to-retail margins, food manufacturers' ability to alter product lines to meet the needs of consumers' changing preferences, and potential shifts in commodity sales toward international buyers.

Moving forward, a host of additional research is needed to further understand the impacts of weight management in the US food industry and in the economy at large. As mentioned previously, short- and long-run impacts of weight loss attempts on food consumption and expenditures should be more precisely evaluated, in addition to within-group consumption changes (e.g., 80 percent to 90 percent lean ground beef within the beef category). An extension to these efforts should be an assessment of food preferences among household members of those attempting to lose weight. That is, do weight loss attempts and corresponding food purchasing decisions have habit-formation effects on, say, children in the household?

More broadly, we have shown that the composition of consumers' diets change as they attempt to lose weight, but more research is needed to quantify the impacts on the US food industry. Intuitively, the economic "pie" for industry shrinks as consumers eat less, but what happens to the size of the "slices?" That is, which food groups or products represent progressively larger or smaller shares of consumers' diets as weight loss is attempted? Our results, for instance, indicate consumption-reducing impacts of weight loss on processed grain products, lessening the food group's share of the diet and share of expenditures. These size-of-the-pie and share-of-the-pie effects create opportunity for some sectors in the food industry, while harming others. Improved understanding of these impacts on specific sectors and products is needed to ensure that providers of food can appropriately mitigate the financial risk of changing consumption patterns while providing consumers with the food products that they most desire. This study lays the foundation for future research and, we hope,

encourages other economists to tackle emerging issues related to health trends, dietary patterns, and food industry economic performance.

Endnote

¹The lowest BMI category, corresponding to underweight individuals, is seldomly reported. As such, it is combined with the intermediate BMI category, corresponding to normal weight individuals, for outcome estimation purposes. Further, BMI can be validly included as a regressor in outcome estimation as prior-day consumption has no effect on broad BMI category (i.e., there is no simultaneity issue), which is determined by months or years of consumption behavior. Additionally, recreational activity is captured as a yes/no response by NHANES participants when asked if their typical week involves those activities. We do not believe prior-day food consumption influences recreational activity that is measured on a historical, weekly basis (i.e., no simultaneity).

References

- Aboulenein, A., and P. Wingrove. 2024. "Wegovy to be Covered by US Medicare for Heart Disease Patients." Reuters. <https://www.reuters.com/business/healthcare-pharmaceuticals/wegovy-be-covered-by-us-medicare-heart-disease-patients-2024-03-21/>.
- Ahluwalia, N., J. Dwyer, A. Terry, A. Moshfegh, and C. Johnson. 2016. "Update on NHANES Dietary Data: Focus on Collection, Release, Analytical Considerations, and Uses to Inform Public Policy." *Advances in Nutrition* 7, no. 1: 121–134. <https://doi.org/10.3945/an.115.009258>.
- Almada, L., I. McCarthy, and R. Tchernis. 2016. "What Can We Learn About the Effects of Food Stamps on Obesity in the Presence of Misreporting?" *American Journal of Agricultural Economics* 98, no. 4: 997–1017. <https://doi.org/10.1093/ajae/aaw017>.
- Austin, P. C., and E. A. Stuart. 2015. "Moving Towards Best Practice When Using Inverse Probability of Treatment Weighting (IPTW) Using the Propensity Score to Estimate Causal Treatment Effects in Observational Studies." *Statistics in Medicine* 34, no. 28: 3661–3679. <https://doi.org/10.1002/sim.6607>.
- Bina, J. D., and G. T. Tonsor. 2024. "Non-Linear Association of Exercise With Animal Protein Consumption among U.S. Adults." *Nutrients* 16, no. 10: 1438. <https://doi.org/10.3390/nu16101438>.
- Bleich, S. N., and J. A. Wolfson. 2014. "Weight Loss Strategies: Association With Consumption of Sugary Beverages, Snacks and Values about Food Purchases." *Patient Education and Counseling* 96, no. 1: 128–134. <https://doi.org/10.1016/j.pec.2014.04.008>.
- Brenan, M. 2024. "43% of Americans Say They Are Overweight; 55% Want to Slim down." Gallup. <https://news.gallup.com/poll/654425/americans-say-overweight-slim-down.aspx>.
- Carlson, A., K. Kuczynski, T. Pannucci, et al. 2020. *Estimating Prices for Foods in the National Health and Nutrition Examination Survey: The Purchase to Plate Price Tool* (Technical Bulletin 1955; Technical Bulletin). USDA Economic Research Service. https://ers.usda.gov/sites/default/files/_laserfiche/publications/99295/TB-1955.pdf?v=42411.
- Carlson, A., E. Page, T. Zimmerman, C. Tornow, and S. Hermansen. 2019. *Linking USDA Nutrition Databases to IRI Household-Based and Store-Based Scanner Data*, (Technical Bulletin 1952; Technical Bulletin). USDA Economic Research Service. https://ers.usda.gov/sites/default/files/_laserfiche/publications/92571/TB-1952.pdf?v=84362.
- Carlson, A. C., C. E. Tornow, E. T. Page, A. Brown McFadden, and T. Palmer Zimmerman. 2022. "Development of the Purchase to Plate Crosswalk and Price Tool: Estimating Prices for the National Health and Nutrition Examination Survey (NHANES) Foods and Measuring the Healthfulness of Retail Food Purchases." *Journal of Food Composition and Analysis* 106: 104344. <https://doi.org/10.1016/j.jfca.2021.104344>.

- Centers for Disease Control and Prevention. 2024a. *Adult BMI Categories*. CDC. <https://www.cdc.gov/bmi/adult-calculator/bmi-categories.html>.
- Centers for Disease Control and Prevention. 2024b. *Measuring Guides for the Dietary Recall Interview*. CDC Archive. https://archive.cdc.gov/www_cdc_gov/nchs/nhanes/measuring_guides_dri/measuringguides.htm.
- Chakraborty, A. 2021. "Present Bias." *Econometrica* 89, no. 4: 1921–1961. <https://doi.org/10.3982/ECTA16467>.
- Chatton, A., and J. M. Rohrer. 2024. "The Causal Cookbook: Recipes for Propensity Scores, G-Computation, and Doubly Robust Standardization." *Advances in Methods and Practices in Psychological Science* 7, no. 1: 1–22. <https://doi.org/10.1177/25152459241236149>.
- Chen, T. C., J. Clark, M. K. Riddles, L. K. Mohadjer, and T. H. I. Fakhouri. 2020. *National Health and Nutrition Examination Survey, 2015–2018: Sample Design and Estimation Procedures*, (2(184); Vital and Health Statistics). National Center for Health Statistics. <https://stacks.cdc.gov/view/cdc/88305>.
- Cochran, W. G. 1968. "The Effectiveness of Adjustment by Subclassification in Removing Bias in Observational Studies." *Biometrics* 24, no. 2: 295–313. <https://doi.org/10.2307/2528036>.
- Dwyer, J., M. F. Picciano, and D. J. Raiten. 2003. "Collection of Food and Dietary Supplement Intake Data: What We Eat in America–NHANES." *Journal of Nutrition* 133, no. 2: 590S–600S. <https://doi.org/10.1093/jn/133.2.590S>.
- Epstein, L. H., and J. J. Leddy. 2006. "Food Reinforcement." *Appetite* 46, no. 1: 22–25. <https://doi.org/10.1016/j.appet.2005.04.006>.
- Epstein, L. H., R. Truesdale, A. Wojcik, R. A. Paluch, and H. A. Raynor. 2003. "Effects of Deprivation on Hedonics and Reinforcing Value of Food." *Physiology & Behavior* 78, no. 2: 221–227. [https://doi.org/10.1016/S0031-9384\(02\)00978-2](https://doi.org/10.1016/S0031-9384(02)00978-2).
- Fan, M., and Y. Jin. 2014. "Obesity and Self-Control: Food Consumption, Physical Activity, and Weight-Loss Intention." *Applied Economic Perspectives and Policy* 36, no. 1: 125–145. <https://doi.org/10.1093/aep/ppt034>.
- Farvid, M. S., E. Sidahmed, N. D. Spence, K. Mante Angua, B. A. Rosner, and J. B. Barnett. 2021. "Consumption of Red Meat and Processed Meat and Cancer Incidence: A Systematic Review and Meta-Analysis of Prospective Studies." *European Journal of Epidemiology* 36, no. 9: 937–951. <https://doi.org/10.1007/s10654-021-00741-9>.
- Federal Reserve Bank of St. Louis. 2024. *Population, Total for United States*. FRED Economic Data. <https://fred.stlouisfed.org/series/POPTOTUSA647NWDB>.
- Food Surveys Research Group. 2024. *Food and Nutrient Database for Dietary Studies*. USDA Agricultural Research Service. <https://www.ars.usda.gov/northeast-area/beltsville-md-bhnrc/beltsville-human-nutrition-research-center/food-surveys-research-group/docs/fndds-download-databases/>.
- Funk, M. J., D. Westreich, C. Wiesen, T. Stürmer, M. A. Brookhart, and M. Davidian. 2011. "Doubly Robust Estimation of Causal Effects." *American Journal of Epidemiology* 173, no. 7: 761–767. <https://doi.org/10.1093/aje/kwq439>.
- Glanz, K., M. Basil, E. Maibach, J. Goldberg, and D. Snyder. 1998. "Why Americans Eat What They Do: Taste, Nutrition, Cost, Convenience, and Weight Control Concerns as Influences on Food Consumption." *Journal of the American Dietetic Association* 98, no. 10: 1118–1126. [https://doi.org/10.1016/S0002-8223\(98\)00260-0](https://doi.org/10.1016/S0002-8223(98)00260-0).
- Greenland, S. 2003. "Quantifying Biases in Causal Models: Classical Confounding vs Collider-Stratification Bias." *Epidemiology* 14, no. 3: 300–306. <https://doi.org/10.1097/01.EDE.0000042804.12056.6C>.
- Grosso, G., L. A. Vignera, R. A. Condorelli, et al. 2022. "Total, Red and Processed Meat Consumption and Human Health: An Umbrella Review of Observational Studies." *International Journal of Food Sciences and Nutrition* 73, no. 6: 726–737. <https://doi.org/10.1080/09637486.2022.2050996>.
- Gundersen, C., B. Kreider, and J. V. Pepper. 2017. "Partial Identification Methods for Evaluating Food Assistance Programs: A Case Study of the Causal Impact of SNAP on Food Insecurity." *American Journal of Agricultural Economics* 99, no. 4: 875–893. <https://doi.org/10.1093/ajae/aa026>.
- Halton, T. L., and F. B. Hu. 2004. "The Effects of High Protein Diets on Thermogenesis, Satiety and Weight Loss: A Critical Review." *Journal of the American College of Nutrition* 23, no. 5: 373–385. <https://doi.org/10.1080/07315724.2004.10719381>.
- Hill, C., J. Saxton, L. Webber, J. Blundell, and J. Wardle. 2009. "The Relative Reinforcing Value of Food Predicts Weight Gain in a Longitudinal Study of 7–10-Y-Old Children." *American Journal of Clinical Nutrition* 90, no. 2: 276–281. <https://doi.org/10.3945/ajcn.2009.27479>.
- Ho, D., K. Imai, G. King, E. Stuart, A. Whitworth, and N. Griefer. 2024. *MatchIt: Nonparametric Preprocessing for Parametric Causal Inference* (Version 4.6.0) [R]. <https://doi.org/10.32614/CRAN.package.MatchIt>.
- Iacus, S. M., G. King, and G. Porro. 2011. "Multivariate Matching Methods That Are Monotonic Imbalance Bounding." *Journal of the American Statistical Association* 106, no. 493: 345–361. <https://doi.org/10.1198/jasa.2011.tm09599>.
- Krinsky, I., and A. L. Robb. 1986. "On Approximating the Statistical Properties of Elasticities." *Review of Economics and Statistics* 68, no. 4: 715–719. <https://doi.org/10.2307/1924536>.
- Latif, W., K. J. Lambrinos, P. Patel, and R. Rodriguez. 2024. *Compare and Contrast the Glucagon-Like Peptide-1 Receptor Agonists (GLP1RAs)*. National Library of Medicine. <http://www.ncbi.nlm.nih.gov/books/NBK572151/>.
- Leidy, H. J., P. M. Clifton, A. Astrup, et al. 2015. "The Role of Protein in Weight Loss and Maintenance." *American Journal of Clinical Nutrition* 101, no. 6: 1320S–1329S. <https://doi.org/10.3945/ajcn.114.084038>.
- Love, D. C., F. Asche, Z. Conrad, et al. 2020. "Food Sources and Expenditures for Seafood in the United States." *Nutrients* 12, no. 6: 1810. <https://doi.org/10.3390/nu12061810>.
- Mayo Clinic. 2024. *Considering GLP-1 Medications? What They Are and Why Lifestyle Change Is Key to Sustained Weight Loss*. Mayo Clinic. <https://diet.mayoclinic.org/us/blog/2023/considering-glp-1-medications-what-they-are-and-why-lifestyle-change-is-key-to-sustained-weight-loss/>.
- McDonald, J. F., and R. A. Moffitt. 1980. "The Uses of Tobit Analysis." *Review of Economics and Statistics* 62, no. 2: 318–321. <https://doi.org/10.2307/1924766>.
- Mishra, S., J. D. Goldman, N. R. Sahyoun, and A. J. Moshfegh. 2017. "Dietary Protein Intake by Meal Types in Adults Aged 51 Years and Over: WWEIA, NHANES 2011–2012." *Journal of Food Composition and Analysis* 64: 93–96. <https://doi.org/10.1016/j.jfca.2017.07.011>.
- Montero, A., G. Sparks, M. Presiado, and L. Hamel. 2024. *KFF Health Tracking Poll May 2024: The Public's Use and Views of GLP-1 Drugs*. KFF. <https://www.kff.org/health-costs/poll-finding/kff-health-tracking-poll-may-2024-the-publics-use-and-views-of-glp-1-drugs/>.
- Morris, M. S., and P. F. Jacques. 2013. "Total Protein, Animal Protein and Physical Activity in Relation to Muscle Mass in Middle-Aged and Older Americans." *British Journal of Nutrition* 109, no. 7: 1294–1303. <https://doi.org/10.1017/S0007114512003133>.
- National Center for Health Statistics. 2024a. *NHANES Questionnaires, Datasets, and Related Documentation*. Centers for Disease Control and Prevention. <https://www.cdc.gov/nchs/nhanes/Default.aspx>.
- National Center for Health Statistics. 2024b. *Weighting*. Centers for Disease Control and Prevention. <https://www.cdc.gov/nchs/nhanes/tutorials/weighting.aspx>.
- O'Connor, L. E., E. A. Wambogo, K. A. Herrick, R. Parsons, and J. Reedy. 2022. "A Standardized Assessment of Processed Red Meat and Processed Poultry Intake in the US Population Aged ≥2 Years Using

- NHANES.” *Journal of Nutrition* 152, no. 1: 190–199. <https://doi.org/10.1093/jn/nxab316>.
- Paddon-Jones, D., E. Westman, R. D. Mattes, R. R. Wolfe, A. Astrup, and M. Westerterp-Plantenga. 2008. “Protein, Weight Management, and Satiety.” *American Journal of Clinical Nutrition* 87, no. 5: 1558S–1561S. <https://doi.org/10.1093/ajcn/87.5.1558S>.
- Pasiakos, S. M., S. Agarwal, H. R. Lieberman, and V. L. Fulgoni. 2015. “Sources and Amounts of Animal, Dairy, and Plant Protein Intake of US Adults in 2007–2010.” *Nutrients* 7, no. 8: 7058–7069. <https://doi.org/10.3390/nu7085322>.
- Pikosky, M. A., C. J. Cifelli, S. Agarwal, and V. L. Fulgoni. 2022. “Association of Dietary Protein Intake and Grip Strength Among Adults Aged 19+ Years: NHANES 2011–2014 Analysis.” *Frontiers in Nutrition* 9: 873512. <https://www.frontiersin.org/articles/10.3389/fnut.2022.873512>.
- Planet Fitness. 2024. *SEC Filings*. Planet Fitness. <https://investor.planetfitness.com/investors/financial-information/sec-filings/default.aspx>.
- Raynor, H. A., and L. H. Epstein. 2003. “The Relative-Reinforcing Value of Food under Differing Levels of Food Deprivation and Restriction.” *Appetite* 40, no. 1: 15–24. [https://doi.org/10.1016/S0195-6663\(02\)00161-7](https://doi.org/10.1016/S0195-6663(02)00161-7).
- Roe, B. E. 2025. “Food Demand in a Post-Ozempic World.” *Choices*, Quarter 1(2025). <https://www.choicesmagazine.org/choices-magazine/submitted-articles/food-demand-in-a-post-ozempic-world>.
- Sievert, K., M. Lawrence, C. Parker, C. A. Russell, and P. Baker. 2022. “Who Has a Beef With Reducing Red and Processed Meat Consumption? A Media Framing Analysis.” *Public Health Nutrition* 25, no. 3: 578–590. <https://doi.org/10.1017/S1368980021004092>.
- Snowden, J. M., S. Rose, and K. M. Mortimer. 2011. “Implementation of G-Computation on a Simulated Data Set: Demonstration of a Causal Inference Technique.” *American Journal of Epidemiology* 173, no. 7: 731–738. <https://doi.org/10.1093/aje/kwq472>.
- Stuart, E. A. 2010. “Matching Methods for Causal Inference: A Review and a Look Forward.” *Statistical Science* 25, no. 1: 1–21. <https://doi.org/10.1214/09-STS313>.
- Tobin, J. 1958. “Estimation of Relationships for Limited Dependent Variables.” *Econometrica* 26, no. 1: 24–36. <https://doi.org/10.2307/1907382>.
- Todd, J., L. Mancino, and B.-H. Lin. 2010. *The Impact of Food Away from Home on Adult Diet Quality*, (90; Economic Research Report). U.S. Department of Agriculture Economic Research Service.
- U.S. Bureau of Labor Statistics. 2024. *American Time Use Survey*. U.S. Bureau of Labor Statistics. <https://www.bls.gov/tus/database/tusa-1tabs.htm>.
- U.S. Department of Agriculture Economic Research Service. 2025. *Farm Income and Wealth Statistics—Cash Receipts by Commodity*. Economic Research Service. <https://data.ers.usda.gov/reports.aspx?ID=4057>.
- U.S. Department of Agriculture Economic Research Service. 2024. *Purchase to Plate*. Economic Research Service. <https://www.ers.usda.gov/data-products/purchase-to-plate/>.
- U.S. Department of Agriculture Food and Nutrition Service. 2025. *SNAP Eligibility*. USDA Food and Nutrition Service. <https://www.fns.usda.gov/snap/recipient/eligibility>.
- U.S. Department of Agriculture. 2025. *What Is MyPlate?*, USDA MyPlate. <https://www.myplate.gov/eat-healthy/what-is-myplate>.
- Vansteelandt, S., and N. Keiding. 2011. “Invited Commentary: G-Computation—Lost in Translation?” *American Journal of Epidemiology* 173, no. 7: 739–742. <https://doi.org/10.1093/aje/kwq474>.
- Vass, C. M., M. Boeri, C. Poulos, and A. J. Turner. 2022. “Matching and Weighting in Stated Preferences for Health Care.” *Journal of Choice Modelling* 44: 100367. <https://doi.org/10.1016/j.jocm.2022.100367>.
- Wang, Y., M. A. Beydoun, B. Caballero, T. L. Gary, and R. Lawrence. 2010. “Trends and Correlates in Meat Consumption Patterns in the US Adult Population.” *Public Health Nutrition* 13, no. 9: 1333–1345. <https://doi.org/10.1017/S1368980010000224>.
- Wardle, J., J. Griffith, F. Johnson, and L. Rapoport. 2000. “Intentional Weight Control and Food Choice Habits in a National Representative Sample of Adults in the UK.” *International Journal of Obesity* 24, no. 5: 534–540. <https://doi.org/10.1038/sj.ijo.0801188>.
- Westreich, D., and S. Greenland. 2013. “The Table 2 Fallacy: Presenting and Interpreting Confounder and Modifier Coefficients.” *American Journal of Epidemiology* 177, no. 4: 292–298. <https://doi.org/10.1093/aje/kws412>.
- Willits-Smith, A., H. Odinga, K. O’Malley, and D. Rose. 2023. “Demographic and Socioeconomic Correlates of Disproportionate Beef Consumption among US Adults in an Age of Global Warming.” *Nutrients* 15, no. 17: 3795. <https://doi.org/10.3390/nu15173795>.
- Witters, D., and E. Maese. 2024a. *Injectable Weight Loss Drugs: Who Uses Them, and Do They Work?*, Gallup. <https://news.gallup.com/poll/644861/injectable-weight-loss-drugs-uses-work.aspx>.
- Wooldridge, J. 2010. *Econometric Analysis of Cross Section and Panel Data*, (2nd ed.). The MIT Press.
- Zeng, L., M. Ruan, J. Liu, et al. 2019. “Trends in Processed Meat, Unprocessed Red Meat, Poultry, and Fish Consumption in the United States, 1999–2016.” *Journal of the Academy of Nutrition and Dietetics* 119, no. 7: 1085–1098. <https://doi.org/10.1016/j.jand.2019.04.004>.
- Zipf, G., M. Chiappa, K. Porter, Y. Ostchega, B. Lewis, and J. Dostal. 2013. *National Health and Nutrition Examination Survey: Plan and Operations, 1999–2010* (1(56); Vital and Health Statistics). National Center for Health Statistics. https://www.cdc.gov/nchs/data/series/sr_01/sr01_056.pdf.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supplementary information: agec70072-sup-0001-Appendix.docx **Supplementary Material:** agec70072-sup-0002-SuppMat.zip